Project Proposal

Smart Emergency Response & Healthcare Coordination Platform

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1. Project Idea

The Smart Emergency Response & Healthcare Coordination Platform is a web-based system designed to improve emergency response by connecting patients, emergency center operators, ambulance teams, hospitals, doctors, and patients' emergency contacts through a centralized platform.

When a patient experiences a medical emergency, they can submit an emergency request containing their location, emergency type, and relevant medical information. The Emergency Center Operator reviews the request and confirms the appropriate ambulance assignment. The platform supports ambulance tracking, hospital coordination, real-time status updates, and notifications to authorized parties until the emergency case is completed.

The system also uses AI-assisted triage to suggest an emergency type, priority level, and suitable medical department based on the information provided. These suggestions support decision-making but do not replace professional medical judgment or the Emergency Center Operator's final decision.

The main goal is to reduce coordination delays, improve communication, and help emergency teams access relevant information before the patient arrives at the hospital.

2. Problem Statement

Emergency response may be delayed when patients cannot clearly communicate their condition, emergency services do not have accurate location information, or ambulance and hospital teams lack timely updates.

In addition, medical staff may not receive important information about a patient's allergies, chronic diseases, or current medications before arrival. Family members or emergency contacts may also have difficulty receiving reliable updates about the patient's situation.

When these activities depend on separate phone calls and disconnected communication channels, coordination becomes more difficult.

Therefore, a centralized platform is needed to coordinate emergency requests, ambulance dispatch, hospital preparation, medical information sharing, and status notifications.

3. Proposed Solution

The proposed platform provides a centralized environment for managing the emergency journey, from submitting an emergency request to hospital treatment and case closure.

The system consists of:

- **Patient Web Application:** Allows patients to manage their profiles, submit emergency requests, and track their cases.
- **Ambulance Web Application:** Allows ambulance teams to receive assignments, access case information, and update their status and location.

- **Emergency Center Dashboard:** Enables operators to review emergency requests, evaluate AI-assisted suggestions, confirm ambulance assignments, and monitor ongoing cases.
- **Hospital Dashboard:** Allows authorized hospital staff to receive incoming case information, review estimated arrival times, and prepare for patient arrival.
- **Doctor Dashboard:** Enables authorized doctors to access relevant medical information and document assessments and treatment notes.
- **Admin Dashboard:** Supports user and system management, monitoring, and reporting.

Emergency Workflow

1. **Registration and Medical Profile:** The patient registers using email or Google Login and may complete an Emergency Medical Profile.
2. **Emergency Request:** The patient submits a request, selects the emergency type, and shares their GPS location when available.
3. **AI-Assisted Triage:** The system suggests the emergency type, priority, and appropriate medical department based on the available information.
4. **Emergency Review:** The Emergency Center Operator reviews the request, patient information, location, and AI suggestions.
5. **Ambulance Assignment:** The system recommends a suitable available ambulance based on relevant factors such as distance, availability, and capabilities. The operator confirms the assignment.
6. **Ambulance Tracking:** The ambulance team accepts the assignment and updates its status as it travels to the patient and then to the hospital.
7. **Hospital Coordination:** The platform recommends a suitable hospital based on factors such as department availability, capacity, distance, and estimated arrival time. The responsible staff member confirms the destination.
8. **Hospital Preparation and Treatment:** Authorized hospital staff receive relevant case information and the estimated arrival time. Doctors access the medical information required for treatment and record relevant notes.
9. **Case Closure:** Authorized staff complete the case workflow, and the system stores its history for monitoring and reporting.

Emergency contacts can receive relevant notifications about the patient's request and status, subject to the system's privacy and access-control rules.

4. Project Objectives

The main objectives are to:

- Provide a simple way for patients to submit emergency requests.
- Capture and share the patient's location when GPS is available.
- Provide an Emergency Center dashboard for reviewing and coordinating emergency cases.
- Recommend suitable ambulances and hospitals using relevant operational information.

- Support real-time updates for ambulance location and emergency status.
- Maintain an Emergency Medical Profile containing relevant medical information.
- Enable authorized medical personnel to access necessary patient information.
- Notify emergency contacts about important case updates.
- Use AI to support emergency triage without replacing human decision-making.
- Provide secure authentication and role-based access control.
- Generate reports about emergency cases, response times, ambulance utilization, and hospital workload.
- Improve communication between patients, emergency services, ambulance teams, and hospitals.

5. Target Users and Roles

The platform defines six main system roles:

Role	Responsibilities
Patient	Manages personal information, maintains an optional medical profile, submits emergency requests, and tracks case status.
Emergency Center Operator	Reviews emergency requests, evaluates available information, confirms ambulance assignments, and coordinates emergency response.
Ambulance / Paramedic	Receives assignments, navigates to the patient, updates ambulance status, and provides relevant handover information.
Hospital Staff	Receives incoming case notifications, reviews relevant information, prepares for arrival, and updates hospital-side case status.
Doctor	Reviews relevant medical information, assesses the patient, and documents treatment notes.
Admin	Manages authorized system users, monitors system activity, and reviews reports.

Emergency Contacts: These are recipients of emergency notifications and relevant status updates. They are not necessarily a separate system role and do not require independent accounts unless such functionality is introduced later.

6. Main Features

6.1 Patient Features

- Registration and login using email or Google Login.
- Personal profile management.
- Emergency Medical Profile containing relevant information such as allergies, chronic diseases, current medications, emergency contacts, and optional blood type.
- Emergency request submission.
- GPS location sharing when available and permitted.

- Selection of an emergency type: chest pain, stroke symptoms, accident, breathing difficulty, severe injury, pregnancy emergency, burn, severe bleeding, or other.
- Viewing emergency status, ambulance location, and estimated arrival time when available.

6.2 Emergency Center Features

- View and manage incoming emergency requests.
- Review the patient's emergency information and location.
- View AI-assisted triage suggestions.
- Confirm, assign, or change the recommended ambulance.
- Monitor ongoing cases and ambulance status.
- Coordinate with ambulance teams and hospitals.
- Trigger relevant notifications to patients and emergency contacts.

6.3 Ambulance Features

- Receive emergency assignment notifications.
- View case location and relevant emergency information.
- Accept assignments and update case progress.
- Update statuses such as En Route to Patient, Arrived at Patient, Patient Picked Up, En Route to Hospital, and Arrived at Hospital.
- Share ambulance location when tracking is enabled.
- Access navigation support.

6.4 Hospital Features

- Receive incoming emergency case notifications.
- View relevant patient information and ambulance estimated arrival time.
- Prepare the appropriate department for the incoming patient.
- Update the hospital's case status.
- Record relevant arrival and handover information.

6.5 Doctor Features

- Access authorized patient medical information relevant to treatment.
- Review available symptoms, medical history, and case details.
- Record initial assessments and treatment notes.
- Update the medical portion of the case according to assigned permissions.

6.6 Admin and Reporting Features

- Manage system users and their roles.
- Monitor emergency requests and overall system activity.

- View total, active, and completed emergency cases.
- Monitor average response time and ambulance utilization.
- Review hospital workload and emergency-type statistics.
- Generate daily, weekly, and monthly reports.

6.7 Platform-Wide Features

- Role-based access control.
- Real-time updates using SignalR where applicable.
- Notifications for relevant users and emergency contacts.
- Emergency status history.
- Map-based location display and ambulance tracking.
- AI-assisted triage recommendations.
- Ambulance and hospital recommendations based on available operational data.

7. AI-Assisted Triage and Decision-Making

The AI component analyzes the information provided in an emergency request to suggest a possible emergency category, priority level, and suitable medical department. Where supported, it may also provide a confidence level for its suggestions.

The AI component is intended to support coordination and prioritization. It does not diagnose medical conditions, guarantee the correctness of a recommendation, or replace professional medical judgment.

The Emergency Center Operator remains responsible for reviewing the request and confirming the operational response. Ambulance and hospital recommendations are also subject to confirmation by the responsible staff members.

8. Security and Patient Data Privacy

Because the platform handles sensitive emergency and medical information, protecting patient data is a core requirement.

The system will apply the following principles:

- Secure authentication and role-based authorization.
- Restrict medical information to authorized personnel who need it for their assigned responsibilities.
- Protect sensitive information during transmission using HTTPS.
- Validate permissions on backend APIs rather than relying only on frontend restrictions.
- Avoid exposing sensitive patient information in application logs or unauthorized notifications.
- Share only the information necessary for each user's role and task.
- Test access permissions to reduce the risk of unauthorized disclosure.

Emergency contacts will receive relevant updates according to the system's notification rules and privacy controls.

9. Minimum Viable Product (MVP)

The first working version will focus on completing the essential emergency workflow before adding advanced functionality.

The MVP will include:

1. Authentication and role-based access control.
2. Patient profile and Emergency Medical Profile.
3. Emergency request creation with location information when available.
4. Emergency Center dashboard for reviewing requests.
5. Ambulance assignment and status updates.
6. Basic hospital coordination and case handover.
7. Relevant notifications and emergency status history.
8. Basic reports for emergency cases and response times.

The basic versions of AI-assisted triage, ambulance and hospital matching, and real-time tracking are part of the core delivery. Advanced improvements, such as more sophisticated AI triage, optimized matching, extensive analytics, and advanced tracking, will be added according to available time and technical feasibility.

10. Project Scope and Limitations

The project focuses on emergency response coordination, ambulance dispatch, hospital preparation, relevant medical information sharing, and emergency case tracking.

The project does not aim to replace doctors, provide definitive medical diagnoses, or make final medical decisions automatically.

Blood bank management is outside the project scope. The optional blood type field in the patient's medical profile does not represent a blood bank feature.

The platform does not depend on government APIs. Any external integration, including Google Login and mapping services, will depend on service availability, access permissions, and technical feasibility.

11. Expected Outcomes

The expected outcomes include:

- Faster coordination between emergency response participants.
- Better visibility of emergency case progress.
- Improved access to relevant patient information for authorized medical staff.
- Better preparation by hospitals before patient arrival.

- More effective communication with patients and their emergency contacts.
- Improved monitoring through dashboards and reports.
- A centralized and traceable emergency workflow.

12. Conclusion

The Smart Emergency Response & Healthcare Coordination Platform aims to improve emergency coordination through a centralized web-based system that connects patients, emergency center operators, ambulance teams, hospitals, and doctors.

By combining emergency request management, location tracking, ambulance and hospital recommendations, AI-assisted triage, secure medical information sharing, and reporting, the platform seeks to support a more coordinated and transparent emergency response process.

The system is designed to assist healthcare professionals and emergency teams, while keeping operational and medical decisions under the responsibility of authorized personnel.